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A compartment model for subcutaneous injection of monoclonal antibodies

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ABSTRACT

Despite the growing popularity of subcutaneous (SC) administration for monoclonal antibodies (mAbs), there remains a limited understanding of the significance of mAb transport rate constants within the interstitial space and the lymphatic system on their pharmacokinetics. To bridge this knowledge gap, we introduce a compartmental model for subcutaneously administered mAbs. Our model differentiates FcRn-expressing cells across various sites, and the model predictions agree with experimental data from both human and rat studies. Our findings indicate that the time to reach the maximum mAb concentration in the plasma, denoted by T_{max}, displays a weak positive correlation with mAb half-life and a negligible correlation with bioavailability. In contrast, the half-life of mAbs exhibits a strong positive correlation with bioavailability. Moreover, the rate of mAb transport from lymph to plasma significantly affects the mAb half-life. Increasing the transport rates of mAbs from the injection site to the lymph or from lymph to plasma enhances bioavailability. These insights, combined with our compartmental model, contribute to a deeper understanding of the pharmacokinetics of subcutaneously administered mAbs.

1. Introduction

Monoclonal antibodies have revolutionized the disease treatment by precisely targeting and binding to disease-associated molecules or cells (Leader et al., 2008). The US Food and Drug Administration has approved 103 therapeutic antibody drugs until 2021, including 34 that are formulated with concentrations greater than 100 mg/mL (Wang et al., 2021). Among these 34 mAbs with high concentrations, 76% are administered through SC route, and 15% are administered intravenously (Wang et al., 2021). The SC injection is relatively easy to administer, minimally invasive, and can be performed by the patient or a caregiver at home (Davies et al., 2014; Sree et al., 2023). Additionally, the SC route allows for a slow release of the therapeutic proteins, which helps to maintain a consistent and stable level in the body. Therefore, understanding the transport of mAbs following subcutaneous injection is crucial for optimizing their efficacy and ensuring optimal therapeutic outcomes.

Experimental studies have demonstrated that protein molecules with a molecular weight greater than 40 kDa are almost completely absorbed via the lymph (Supersaxo et al., 1990; Richter and Jacobsen, 2014). Therefore, large molecules such as mAbs, which have a molecular weight of 150 kDa, are initially absorbed by the initial lymphatics before being transported through the lymphatic system and eventually reaching the systemic circulation (Viola et al., 2018; McLennan et al., 2005). Subcutaneously administered mAbs undergo pre-systemic elimination, resulting in a bioavailability range of 50% to 80% (Richter et al., 2012; Richter and Jacobsen, 2014). Additionally, for most approved subcutaneously administered therapeutic mAbs in the US, the time (T_{max}) to reach their peak concentration in the plasma typically ranges from 1 to 14 days (Ryman and Meibohm, 2017), with most falling between 2 and 8 days (Viola et al., 2018; Zou et al., 2021).

Several compartment models have been developed to predict the transport of mAbs after SC administration (Garg, 2007; Golay et al., 2013; Sánchez-Félix et al., 2020; Kagan, 2014; Hu and D'Argenio, 2020; Glassman and Balthasar, 2016). For instance, Sánchez-Félix et al. (2020) introduced a compartment model that considers the degradation of mAbs due to precipitation at the injection site, as well as an exchange between mAbs in the SC tissue at the injection site and endosomal compartments present in the local SC tissue, vascular, and lymphatic endothelial cells. Golay et al. (2013) presented a compartment model that includes the loss of mAbs from the SC injection site due to degradation and phagocytosis, but it neglects the intracellular catabolism of mAbs in lymphatic vessels. Moreover, Kagan et al. (2012) conducted an experimental study on the role of dose, injection site, and injection volume on the pharmacokinetics of rituximab in rats. They found that bioavailability following subcutaneous injection was inversely related to the dose level due to saturable binding. Additionally, they explored three different compartment models and compared the model predictions against their experimental data. They found that the following

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compartment model can predict the experimental observations: (1) the drug administered at the injection site is transported to the absorption site following the first-order kinetics, (2) at the absorption site, free rituximab either instantly binds to receptors or undergoes proteolytic degradation, (3) both free rituximab and rituximab–receptor complexes are absorbed into the systematic circulation, (4) the systemic disposition of rituximab includes nonspecific distribution and first-order elimination (Kagan et al., 2012).

The neonatal Fc receptor (FcRn) protects mAbs from lysosomal degradation by forming FcRn–mAb complexes (Latvala et al., 2017; Ovacik and Lin, 2018; Li and Balthasar, 2018). This process leads to increased bioavailability when FcRn-expressing cells are present at the injection site and in the lymphatic system, and prolonged half-life when FcRn-expressing cells are present in the plasma (Ghetie et al., 1996; Israel et al., 1996; Keizer et al., 2010). Experiments have demonstrated that the bioavailability of subcutaneously administered IgG1 is three times higher in mice with FcRn compared to FcRn-deficient mice (Wang et al., 2008). However, most current compartment models do not distinguish FcRn-expressing cells at different sites, potentially introducing bias in understanding the role of FcRn-expressing cells in mAbs pharmacokinetics and leading to inaccuracies in model predictions.

In this paper, we present a compartment model that considers FcRn-expressing cells in local SC tissue, lymph, plasma, and peripheral tissues. We assume that there is an adequate number of FcRn-expressing cells in plasma and peripheral tissues, while their quantity is limited in local SC tissue and lymph. Through fitting experimental data on mAb concentration in the plasma for rats (Kagan et al., 2012), we find that the calibrated mass of FcRn in local SC tissue and lymph is approximately 1 mg. Consequently, as the administered therapeutic mAb mass increases from 0.38 to 1.52 mg, the amount of free mAbs susceptible to degradation in FcRn-expressing cells within local SC tissue and lymph increases, resulting in decreased bioavailability. In contrast, when we fit experimental data on mAb concentration in the plasma for humans (Viola et al., 2018), the calibrated mass of FcRn in local SC tissue and lymph is in the magnitude of 0.1 mg, which is significantly smaller than the administered therapeutic mAbs, which is of the order of 100 mg. Therefore, the recycling of mAbs in FcRn-expressing cells within local SC tissue and lymph can be neglected.

Moreover, we employ sensitivity analyses to identify critical factors that impact mAb pharmacokinetics and conduct parametric studies to quantify their impact. Our sensitivity analysis of human data indicates that the cellular uptake rate significantly affects $T_{\max}$, the half-life of mAbs, and their bioavailability. In comparison to the transport rate of therapeutic mAbs from the injection site to the lymph, the rate from lymph to plasma has less influence on $T_{\max}$, yet it has a greater impact on the mAb half-life.

This paper is structured as follows. Section 2 presents the physiology involved in the transport of mAbs after SC injection. Our compartment model is provided in Section 3, and the calibration and validation results are included in Section 4. Section 5 presents our sensitivity analysis and correlation analysis on both human and rat data. The limitations of our model are discussed in Section 6. The concluding remarks are provided in Section 7.

2. Transport of mAbs after SC injection

After SC injection, mAbs in the interstitial space migrate towards the initial lymphatics to enter the lymphatic system. The fluid velocity in the interstitial space ranges from 0.1 to 2 μm/s under normal conditions (Swartz and Fleury, 2007). The diffusion of mAbs in the interstitial space is significantly reduced compared to their diffusion in free solution, with reductions ranging from 20% to 90% depending on various factors such as the density and size of collagen fibers, molecular size of mAbs, and ionic interactions between mAbs and the extracellular matrix (ECM) (Swartz and Fleury, 2007).

The ECM components include proteoglycans (PGs), collagen, elastin, laminin, and fibrin. PGs are large brush-shaped macromolecules composed of a protein core and negatively charged glycosaminoglycan (GAG) chains (Swartz and Fleury, 2007; Frantz et al., 2010). The negative charge of GAGs attracts counterions to preserve electroneutrality, resulting in high interstitial osmotic pressure that influences the diffusion, convection, and binding of molecules and particles within the interstitial space (Swartz, 2001). Studies have shown that protein molecules with higher charge and hydrophobic interactions with the ECM have lower bioavailability (Datta-Mannan et al., 2020).

mAbs can enter the initial lymphatics through either transcellular (Triacca et al., 2017) or paracellular pathways (Salgado and Cao, 2021; Moore and Bertram, 2018). The initial lymphatics are blind-ended structures with diameters of 10–60 μm (Swartz, 2001), and they comprise a single layer of endothelial cells anchored to the surrounding ECM through elastic fibers (Reddy and Patel, 1995). These fibers exert radial tension on the initial lymphatics, acting as a tissue pump to promote lymph formation (Schmid-Schönbein, 1990). The absorption of fluid and mAbs by the initial lymphatics is regulated by the local interstitial pressure, tension in the surrounding ECM, and downstream pressure. The absorbed lymph and mAbs are transported to the collecting lymphatics and pass through at least one lymph node before entering the venous circulation (Moore and Bertram, 2018).

The collecting lymphatics, surrounded by smooth muscles that contract spontaneously in waves traveling at approximately 5 mm/s (Moore and Bertram, 2018), are equipped with valves ensuring unidirectional lymph flow (Swartz, 2001). In normal conditions, neighboring valves do not open or close simultaneously, resulting in stepwise changes in pressure from one lymphangion to the next (Zawieja, 2009). The lymph node is densely packed, adding resistance to the flow (Willard-Mack, 2006; Ikomi et al., 2012). Additionally, the lymph node is connected to blood vessels, allowing the exchange of water, proteins, and cells between lymph and blood (Jafarnejad et al., 2015; Moore and Bertram, 2018). Past numerical studies have suggested that around 10% of the incoming lymph flow is absorbed by the blood vessels (Jafarnejad et al., 2015).

After reaching the venous circulation, mAbs, along with the oxygen-poor blood, are transported to the lungs, where the blood gains oxygen and eliminates waste such as carbon dioxide. Subsequently, mAbs in the arterial blood are distributed to various tissues, including bones, adipose, brain, heart, kidney, muscle, skin, live, spleen, gut, and pancreas (Shah and Betts, 2012; Offman and Edginton, 2015; Gill et al., 2016a). After that, mAbs in the interstitial space are transported to lymph and enter the venous circulation, repeating this cycle. It should be noted that the venous blood from the small intestine, large intestine, spleen, and pancreas is delivered to the liver through the hepatic portal vein (Shah and Betts, 2012).

The elimination of mAbs primarily occurs through intracellular catabolism via lysosomal degradation to amino acids after uptake, either through receptor-mediated endocytosis or pinocytosis (Ryman and Meibohm, 2017). Target-mediated drug disposition (TMDD) refers to the elimination process that receptor-mediated endocytosis is triggered by the interactions of the antigen-binding fragments of the therapeutic mAbs with their pharmacological targets. This target-mediated clearance decreases with the saturation of the target, which, in turn, is dependent on doses (Ovacik and Lin, 2018). Catabolic degradation of mAbs following pinocytotic uptake occurs throughout the body, particularly in organs and tissues rich in capillary beds with endothelial cells. Consequently, the skin, muscles, and gastrointestinal tract are the major elimination organs for mAb molecules that do not undergo receptor-mediated elimination pathways (Ryman and Meibohm, 2017).

FcRn in vascular endothelial cells plays a crucial role in salvaging or transcytosis of administered mAbs (Richter et al., 2018). FcRn exhibits low affinity for mAbs at neutral pH (approximately 7.4). However, once an early endosome forms and undergoes acidification (pH 5–6.5), the affinity of FcRn increases, facilitating the Fc region of the mAbs binding

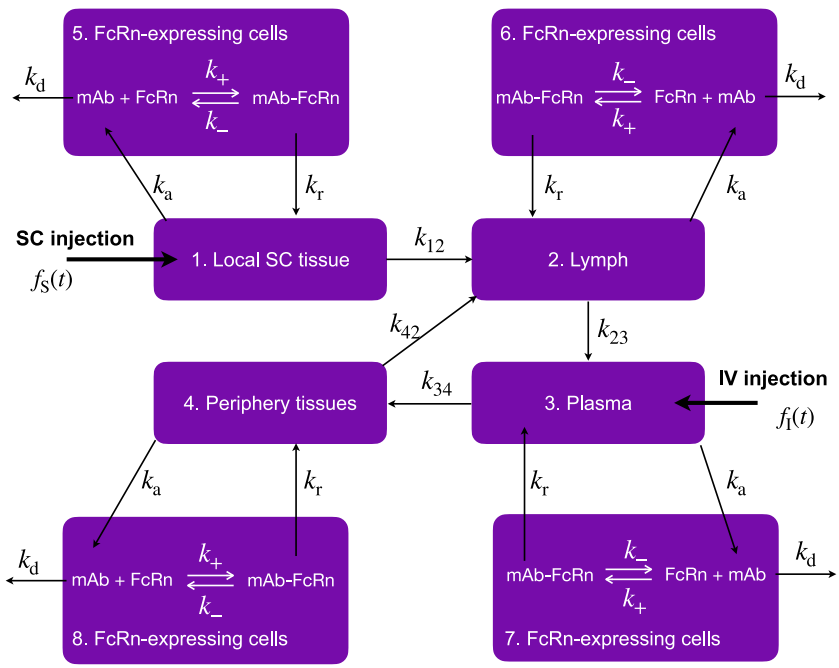

Fig. 1. Our compartment model describing the transport of therapeutic mAbs following subcutaneous or intravenous administration.

to FcRn within endosomes. Subsequently, the FcRn–mAb complex is transported back to the cell surface, where mAbs are released at neutral pH. Meanwhile, proteins in the endosomes that are not bound to FcRn undergo proteolytic degradation in the lysosome (pH < 5) (Ryman and Meibohm, 2017; Pyzik et al., 2023). It is noteworthy that FcRn is widely present throughout various human body tissues, including epithelial, stromal, and parenchymal cells, as well as hematopoietic cells derived from the myeloid lineage (Akilesh et al., 2007; Latvala et al., 2017; Richter et al., 2018). A recent experimental study demonstrated the importance of FcRn in radiosensitive hematopoietic cells for achieving high bioavailability (close to 90%) in subcutaneous injection in mice (Richter et al., 2018). Therefore, when constructing a compartment model for SC administration of therapeutic mAbs, the presence of hematopoietic cells (e.g., macrophages, dendritic cells, and monocytes) in the SC tissue, lymphatics, and plasma should be considered.

3. Compartment model

Fig. 1 shows our compartment model, which distinguishes FcRn-expressing cells at the injection site, lymph, plasma, and peripheral tissues. The rates of absorption and recycling of mAbs in FcRn-expressing cells are assumed to be the same at different sites. Additionally, we assume that the binding of free mAbs to FcRn in the FcRn-expressing cells occurs rapidly; thus, the concentrations of free mAbs and mAb–FcRn complex remain at equilibrium (Kagan et al., 2012).

The eight compartments in Fig. 1 are labeled as 1, 2, 3, 4, 5, 6, 7, and 8, respectively. We use the notation m_i, where $i = 1,2,3,4$, to denote the mass of mAbs in the ith compartment. Additionally, we use $m_{j,\mathrm{f}}$ and $m_{j,\mathrm{b}}$, where $j = 5,6,7,8$, to represent the masses of free mAb and bound mAbs in the jth compartment, respectively. The sum of free and bound mAbs in the jth compartment is denoted by $m_{j,\mathrm{t}}$, satisfying $m_{j,\mathrm{t}} = m_{j,\mathrm{f}} + m_{j,\mathrm{b}}$.

The mass of mAbs in ith compartment, where $i = 1,2,3,4$, is governed by

$$\frac{\mathrm{d}m_1}{\mathrm{d}t} = k_\mathrm{r}\, m_{5,\mathrm{b}} - (k_\mathrm{a} + k_{12})m_1 + f_\mathrm{s}(t) \tag{1}$$

$$\frac{\mathrm{d}m_2}{\mathrm{d}t} = k_{12}\, m_1 + k_\mathrm{r}\, m_{6,\mathrm{b}} + k_{42}\, m_4 - (k_\mathrm{a} + k_{23})m_2 \tag{2}$$

$$\frac{\mathrm{d}m_3}{\mathrm{d}t} = k_{23}\, m_2 + k_\mathrm{r}\, m_{7,\mathrm{b}} - (k_\mathrm{a} + k_{34})m_3 + f_\mathrm{I}(t) \tag{3}$$

$$\frac{\mathrm{d}m_4}{\mathrm{d}t} = k_{34}\, m_3 + k_\mathrm{r}\, m_{8,\mathrm{b}} - (k_\mathrm{a} + k_{42})m_4 \tag{4}$$

where t is the elapsed time after subcutaneous administration; k_{pq} represents the rate of transport of mAbs from the pth compartment to the qth compartment, where $p, q = 1,2,3,4$; variables k_a and k_r denote the rates of absorption and recycling of mAbs in FcRn-expressing cells, respectively; $f_\mathrm{S}(t)$ and $f_\mathrm{I}(t)$ are the rate of subcutaneous injection and intravenous injection, respectively. In the case of administering a single dose therapeutic mAbs instantly into the subcutaneous tissue at $t = 0$, we have $f_\mathrm{I}(t) = 0$ and $f_\mathrm{S}(t) = M\delta(t)$, where M is the mass of therapeutic mAbs, and $\delta(t)$ is the Dirac delta function. The value of $\delta(t)$ is zero everywhere except at $t = 0$, and its integral over t from zero to infinity is equal to one.

In the jth compartment, where $j = 5,6,7,8$, the sum of masses of free and bound mAbs is governed by

$$\frac{\mathrm{d}m_{j,\mathrm{t}}}{\mathrm{d}t} = k_\mathrm{a}\, m_{j-4} - k_\mathrm{d}\, m_{j,\mathrm{f}} - k_\mathrm{r}\, m_{j,\mathrm{b}} \tag{5}$$

where k_d is the degradation rate of free mAbs. The concentrations of free mAbs and mAb–FcRn complexes remain at equilibrium; therefore, we have

$$k_+\, m_{j,\mathrm{f}}\, m_{j,\mathrm{F}} = k_-\, m_{j,\mathrm{b}}, \qquad j = 5,6,7,8 \tag{6}$$

where $m_{j,\mathrm{F}}$ is the instantaneous mass of FcRn in the jth compartment; and k_+ and k_- are association and dissociation rate constants, respectively. Following the literature (Kagan et al., 2012), we assume that the total amount of FcRn in the jth compartment remains constant with time. Therefore, the instantaneous mass of FcRn in the jth compartment is

$$m_{j,\mathrm{F}} = m_{j,\mathrm{F}}\Big|_{t=0} - m_{j,\mathrm{b}}, \qquad j = 5,6,7,8 \tag{7}$$

where $m_{j,\mathrm{F}}\Big|_{t=0}$ is the mass of FcRn in the jth compartment at $t = 0$. Substituting Eq. (7) into Eq. (6), we have

$$m_{j,\mathrm{f}} = \frac{1}{2}\left[m_{j,\mathrm{t}} - m_{j,\mathrm{F}}\Big|_{t=0} - k_\mathrm{diss} + \sqrt{\left(m_{j,\mathrm{F}}\Big|_{t=0} - m_{j,\mathrm{t}} + k_\mathrm{diss}\right)^2 + 4k_\mathrm{diss}\, m_{j,\mathrm{t}}}\right] \tag{8}$$

where $k_\mathrm{diss} = k_-/k_+$ is the equilibrium dissociation constant (Garg and Balthasar, 2007; Kagan et al., 2012).

Note that in the j−th compartment, the mass of mAb–FcRn is smaller than $m_{j,\mathrm{F}}\big|_{t=0}$ and the mass (M) of subcutaneously administered mAbs. If $m_{j,\mathrm{F}}\big|_{t=0} \gg M$, according to Eq. (6), the mass of free mAbs, denoted as $m_{j,\mathrm{f}}$, would be tiny; hence, the j−th compartment can be regarded as a recycling site with negligible degradation. In contrast, if $m_{j,\mathrm{F}}\big|_{t=0} \ll M$, then $m_{j,\mathrm{b}}/m_{j,\mathrm{t}} \approx 0$, and $m_{j,\mathrm{f}}/m_{j,\mathrm{t}} \approx 1$. In this situation, the amount of mAbs recycled from the j−th compartment to the $(j-4)$-th compartment is negligible; hence, the j−th compartment can be considered a degradation site. If the condition either $m_{j,\mathrm{F}}\big|_{t=0} \ll M$ or $m_{j,\mathrm{F}}\big|_{t=0} \gg M$ is satisfied in the j−th compartment, where $j = 5, 6, 7, 8$, then the governing Eqs. (1)–(5) become linear, and the concentration of therapeutic mAbs in the plasma would be proportional to the administered dose.

We consider an abundance of FcRn in the seventh and eighth compartments, i.e., $m_{7,\mathrm{f}} \approx 0$ and $m_{8,\mathrm{f}} \approx 0$. Additionally, we assume limited FcRn masses in the fifth and sixth compartments, with the total FcRn mass in the fifth compartment equal to that in the sixth compartment. This assumption can be expressed as

$$m_{5,\mathrm{F}}\big|_{t=0} = m_{6,\mathrm{F}}\big|_{t=0} = m_{\mathrm{F0}} \tag{9}$$

where m_{F0} is a constant. Governing Eqs. (1)–(5) follow the initial conditions

$$m_i = m_{j,\mathrm{t}} = 0 \qquad \text{at } t = 0 \tag{10}$$

where $i = 1, 2, 3, 4$, and $j = 5, 6, 7, 8$.

4. Model calibration and validation

4.1. Model calibration method

In our compartment model, there are nine unknown parameters: k_{12}, k_{23}, k_{34}, k_{42}, k_{a}, k_{r}, k_{d}, k_{diss}, and m_{F0}. To determine the values of these unknown parameters, we perform model calibration by fitting the experimental measurements of the mass of therapeutic mAbs in the plasma, denoted as $\hat{m}_3$, at various time points $t = t_1, t_2, \ldots, t_N$. For calibration, we employ the following loss function:

$$L = \frac{1}{N}\sum_{p=1}^{N}\left[\log_{10}\hat{m}_3(t_p) - \log_{10} m_3(t_p)\right]^2 \tag{11}$$

where $\hat{m}_3$ and m_3 are the masses of therapeutic mAbs in the plasma measured from experiments and predicted by our compartment model, respectively. Unless otherwise stated, the rate constants k_{12}, k_{23}, k_{34}, k_{42}, k_{a}, k_{r}, and k_{d} have units of day^{-1}, and k_{diss} and m_{F0} have units of milligram.

We utilize the Python package, namely pymcmcstat (Miles and Smith, 2019), which employs Markov Chain Monte Carlo methods with sampling-based Metropolis algorithms (Metropolis et al., 1953) to perform Bayesian calibration. The user-friendly open-source code, pymcmcstat, requires users only to provide loss functions and define model parameters and settings. In our calibration simulations, we assign the ranges [0, 5] to the unknown parameters, k_{12}, k_{23}, k_{34}, k_{42}, k_{a}, k_{r}, k_{d}, and k_{diss}, and set their initial values to 0.5. For the unknown parameter m_{F0}, which represents the mass of FcRn at $t = 0$ in the fifth and sixth compartments, we use the range [0, 100] and set its initial value to 10.

4.2. Calibration and validation for rat data

Kagan et al. (2012) conducted experiments in which rituximab was administered to Wistar rats weighing between 375 and 400 g via intravenous and SC routes, and the concentration of rituximab in plasma was monitored over time. The measured bioavailability for subcutaneous administration of rituximab with doses 1, 10, 40 mg/kg at the back of rats were approximately 70%, 31%, and 18%, respectively (Kagan et al., 2012). We use the software GetData Graph Digitizer to extract data from the figures in the literature (Kagan et al., 2012). In our simulations, the mass of Wistar rats is estimated as 380 g. We utilize two sets of intravenous injection data (1 and 10 mg/kg doses) and two sets of SC injection data (1 and 40 mg/kg doses injected at the back) to calibrate our compartment model. The corresponding predicted means and standard deviations are listed in Table 1. The calibrated mass of FcRn in local SC tissue and lymph is 1.03 mg. Consequently, as the administered therapeutic mAb mass increases from 0.38 to 1.52 mg, the amount of free mAbs susceptible to degradation in FcRn-expressing cells within local SC tissue and lymph increases, leading to a reduction in bioavailability.

After the calibration, we validate our model predictions against another set of SC injection data (10 mg/kg dose injected at the back). The calibration and validation results are shown in Fig. 2. Additionally, our model predictions with 95% confidence intervals for the bioavailability of subcutaneous administration with doses 1, 10, and 40 mg/kg are [48%, 61%], [37%, 52%], and [23%, 35%], respectively. The predicted means for the bioavailability of subcutaneous administration with doses 1, 10, and 40 mg/kg are 55%, 44%, 29%, respectively. The bioavailability is calculated by

$$\text{Bioavailability} = \frac{(\mathrm{AUC})_{\mathrm{sc}} \times (\mathrm{Dose})_{\mathrm{iv}}}{(\mathrm{AUC})_{\mathrm{iv}} \times (\mathrm{Dose})_{\mathrm{sc}}} \tag{12}$$

where the subscripts "sc" and "iv" refer to the results obtained through SC and intravenous injection, respectively. The area under the concentration–time curve (AUC) from time zero to infinity describes the mAbs exposure and can be calculated as

$$\mathrm{AUC} = \int_0^{\infty} C\,\mathrm{d}t. \tag{13}$$

4.3. Calibration and validation for human data

The literature (Varkhede and Forrest, 2018) provides human data for the SC administration of ten therapeutic mAbs, which can be used to validate our compartment model. This data is accessible through the University of Kansas Scholar Works Repository with the link https://kuscholarworks.ku.edu/handle/1808/26648. For repeated experiments, we average the data curves and indicate the standard deviation using error bars. The plasma volume in a 70 kg adult is approximately 3.5 L (Mallick and Bodenham, 2006). Consequently, when calculating the concentration of mAbs in human plasma, we employ a plasma volume of 3.5 L.

For adalimumab, tocilizumab, and infliximab, only single-dose data is available for each of them. For these three mAbs, we fit our model to their respective data. The fitted results are listed in Table 1, and the comparison results are shown in Fig. 3. As for the remaining seven therapeutic mAbs, each of them has multiple-dose data. For each of these seven mAbs, we employ its a subset of data for calibration while keeping data involving distinct doses for validation. For instance, data is available for anifrolumab with doses of 300 and 600 mg. We utilize the canakinumab data at 300 mg for calibration and obtain the values for the unknown parameters k_{12}, k_{23}, k_{34}, k_{42}, k_{a}, k_{r}, k_{d}, k_{diss}, and m_{F0}. Subsequently, we validate our model predictions using the canakinumab data at 600 mg. The calibrated values for these seven therapeutic mAbs are included in Tables 2 and 3, and the comparison results are presented in Figs. 4 and 5.

For human data, the calibrated values for m_{F0} range from 0.01 to 0.1 mg, which is much smaller than the injected dose of mAbs. This suggests that the amount of mAb–FcRn complexes in the fifth and sixth compartments is negligible, as analyzed in Section 3. In other words, once therapeutic mAbs are absorbed into the fifth and sixth compartments, they are trapped with only a minimal portion being recycled back to the first and second compartments. Therefore, when utilizing our compartment model to analyze data from human studies, we can simplify by disregarding the fifth and sixth compartments and

Table 1
The calibration results (predicted mean, standard deviation) for rituximab in the rat study (Kagan et al., 2012) and for adalimumab, tocilizumab, and infliximab in the human study (Varkhede and Forrest, 2018).

Variables	Rituximab	Adalimumab	Tocilizumab	Infliximab	Units
k_{12}	(1.498, 0.003)	(0.714, 0.012)	(1.317, 0.152)	(0.374, 0.043)	day^{-1}
k_{23}	(0.72, 0.014)	(0.312, 0.009)	(0.614, 0.101)	(2.633, 0.303)	day^{-1}
k_{34}	(0.689, 0.014)	(0.212, 0.016)	(0.540, 0.035)	(3.009, 0.618)	day^{-1}
k_{42}	(1.274, 0.016)	(1.024, 0.045)	(3.575, 1.396)	(2.424, 0.436)	day^{-1}
k_a	(2.680, 0.124)	(0.130, 0.006)	(0.406, 0.028)	(0.102, 0.025)	day^{-1}
k_r	(1.796, 0.052)	(0.824, 0.008)	(3.551, 0.969)	(2.228, 0.418)	day^{-1}
k_d	(0.984, 0.028)	(0.377, 0.019)	(0.184, 0.248)	(1.642, 0.588)	day^{-1}
k_{diss}	(0.257, 0.021)	(0.859, 0.007)	(1.871, 0.538)	(0.743, 0.347)	mg
m_{F0}	(1.031, 0.119)	(0.011, 0.001)	(0.019, 0.012)	(0.626, 0.350)	mg

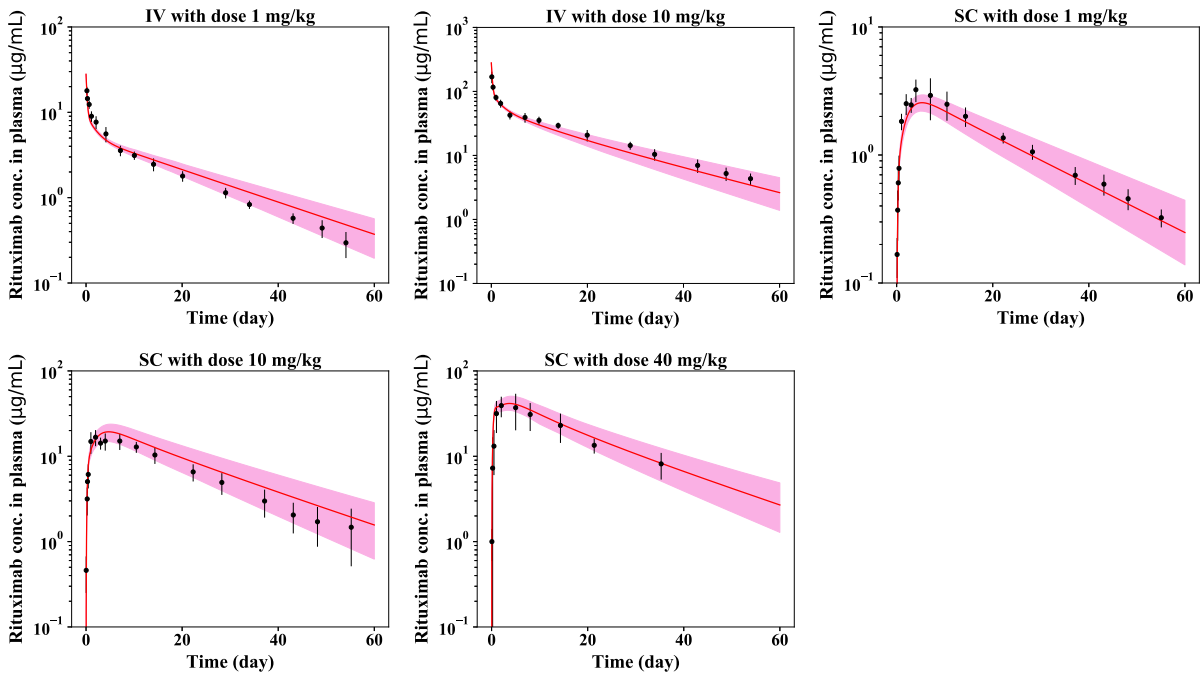

Fig. 2. Comparison between our model predictions and experimental data of injecting rituximab into rats. Solid lines and shaded regions are the model predictions with the mean values of calibrated parameters and 95% confidence intervals, respectively. Dots with error bars are experimental measurements (Kagan et al., 2012). The mass of Wistar rats is estimated as 380 g. The plasma volume of Wistar rats is estimated as 13.6 mL, which is calculated by the empirical formula: plasma volume (mL) = 0.0291 × body weight (g) + 2.54 (Bijsterbosch et al., 1981).

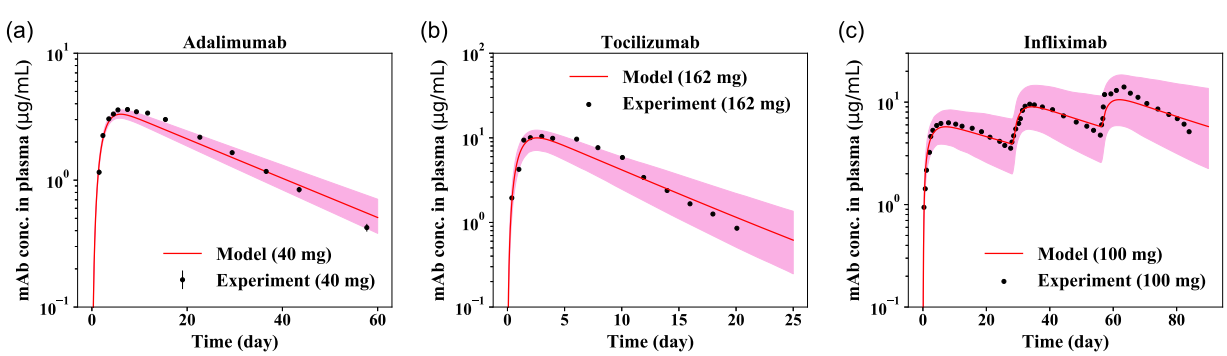

Fig. 3. Model fitting of experimental data for adalimumab, tocilizumab, and infliximab in human studies (Varkhede and Forrest, 2018). Solid line: model predictions with the mean values of calibrated parameters. Shaded regions: model predictions with 95% confidence intervals. Dots: experimental measurements (Varkhede and Forrest, 2018). 100 mg infliximab was administered subcutaneously on days 0, 28, and 56 in figure (c).

instead treating k_a as a degradation rate. Furthermore, the governing Eqs. (1)–(5) display an approximate linearity, indicating an almost linear relationship between the administered doses and the concentration of subcutaneously administered mAbs in the plasma.

The calculated bioavailability and the percentage of degraded mAbs in the 5th and 6th compartments are listed in Table 4. We conduct a correlation analysis among the calibration parameters across the ten mAbs used in the human studies, as listed in Tables 1–3. Their predicted mean values are analyzed, and the results are shown in Figure S1 within the supplementary material. The strength of the relationship between two variables is typically classified as strong, moderate, weak, or negligible when the absolute value of their Pearson correlation coefficient is greater than 0.7, between 0.4 and 0.7, between 0.1 and 0.4, and smaller than 0.1, respectively (Akoglu, 2018). We observe

Table 2
The calibration results (predicted mean, standard deviation) for human studies of SC injection of anifrolumab in healthy volunteers, belimumab in healthy volunteers, belimumab in patients, and canakinumab in healthy volunteers, respectively (Varkhede and Forrest, 2018).

Variables	Anifrolumab	Belimumab (healthy)	Belimumab (patient)	Canakinumab	Units
k_{12}	(1.435, 0.053)	(1.455, 0.040)	(1.405, 0.077)	(0.481, 0.025)	day^{-1}
k_{23}	(1.416, 0.148)	(0.882, 0.038)	(0.423, 0.027)	(0.568, 0.023)	day^{-1}
k_{34}	(0.588, 0.093)	(0.453, 0.010)	(0.159, 0.019)	(0.270, 0.021)	day^{-1}
k_{42}	(3.855, 0.388)	(4.792, 0.175)	(1.544, 0.181)	(1.689, 0.047)	day^{-1}
k_a	(0.227, 0.027)	(0.160, 0.009)	(0.153, 0.016)	(0.113, 0.01)	day^{-1}
k_r	(2.010, 0.080)	(0.830, 0.107)	(0.688, 0.074)	(0.913, 0.038)	day^{-1}
k_d	(0.485, 0.291)	(1.030, 0.023)	(0.960, 0.123)	(0.249, 0.022)	day^{-1}
k_{diss}	(1.472, 0.103)	(0.178, 0.064)	(0.346, 0.180)	(0.481, 0.025)	mg
m_{F0}	(0.038, 0.028)	(0.060, 0.036)	(0.193, 0.114)	(0.023, 0.010)	mg

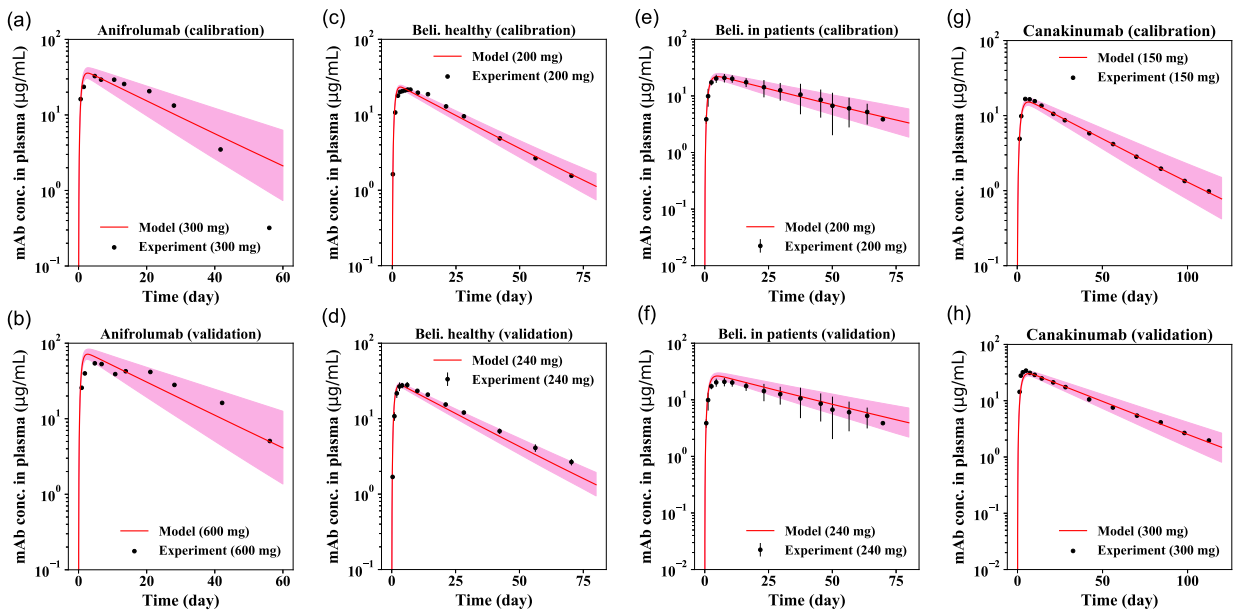

Fig. 4. Calibration and validation results for human studies of SC injection of anifrolumab, canakinumab, belimumab in healthy volunteers, and belimumab in patients (Varkhede and Forrest, 2018). Solid line: model predictions with the mean values of calibrated parameters. Shaded regions: model predictions with 95% confidence intervals. Dots: experimental measurements (Varkhede and Forrest, 2018).

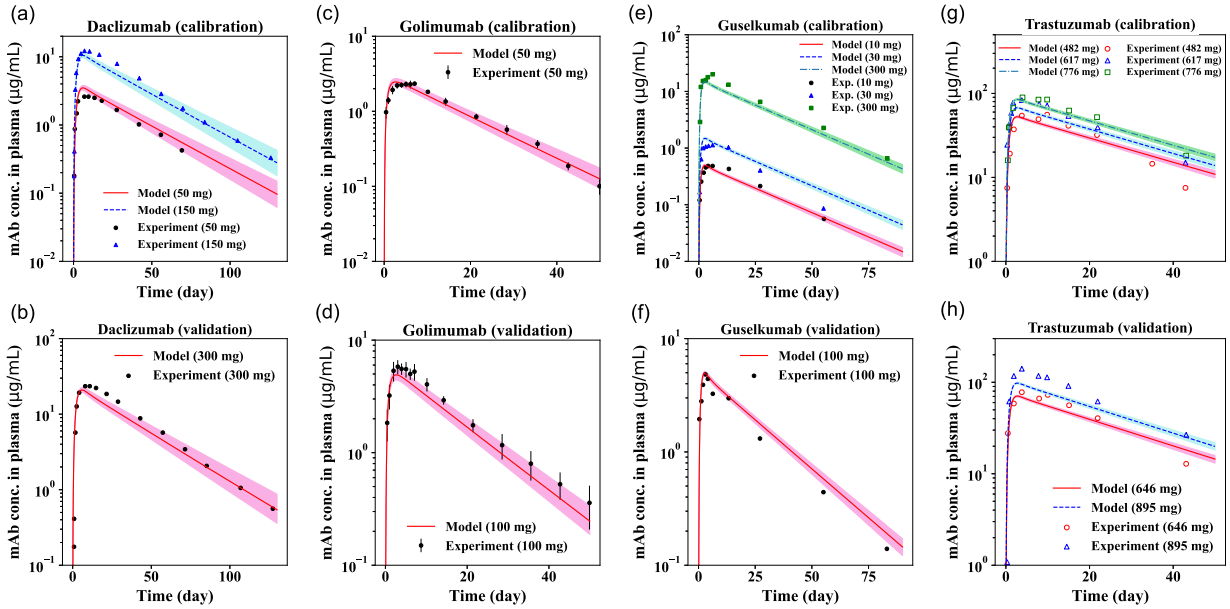

Fig. 5. Calibration and validation results for human studies of SC injection of daclizumab, golimumab, guselkumab, and trastuzumab (Varkhede and Forrest, 2018). Solid line: model predictions with the mean values of calibrated parameters. Shaded regions: model predictions with 95% confidence intervals. Dots: experimental measurements (Varkhede and Forrest, 2018).

Table 3
The calibration results (predicted mean, standard deviation) for human studies of SC injection of daclizumab, golimumab, guselkumab, and trastuzumab, respectively (Varkhede and Forrest, 2018).

Variables	Daclizumab	Golimumab	Guselkumab	Trastuzumab	Units
k_{12}	(0.549, 0.012)	(1.104, 0.031)	(1.492, 0.008)	(1.499, 0.001)	day^{-1}
k_{23}	(0.303, 0.003)	(1.284, 0.055)	(0.408, 0.008)	(1.036, 0.014)	day^{-1}
k_{34}	(0.202, 0.012)	(3.307, 0.196)	(0.146, 0.002)	(0.232, 0.005)	day^{-1}
k_{42}	(0.686, 0.011)	(4.618, 0.295)	(0.812, 0.009)	(1.312, 0.026)	day^{-1}
k_a	(0.134, 0.007)	(0.114, 0.003)	(0.471, 0.005)	(0.311, 0.006)	day^{-1}
k_r	(0.377, 0.025)	(1.517, 0.054)	(0.743, 0.009)	(1.232, 0.023)	day^{-1}
k_d	(0.932, 0.023)	(3.303, 0.176)	(1.428, 0.014)	(0.291, 0.013)	day^{-1}
k_{diss}	(0.737, 0.035)	(0.503, 0.027)	(2.052, 0.015)	(0.294, 0.013)	mg
m_{F0}	(0.117, 0.014)	(0.042, 0.025)	(0.020, 0.008)	(0.087, 0.030)	mg

Table 4
Th predicted mean of SC bioavailability and the percentage of mAbs degraded in the 5th and 6th compartments given by our compartment model.

mAbs	Bioavailability	5th compartment	6th compartment
Adalimumab	60%	15%	85%
Tocilizumab	47%	23%	77%
Anifrolumab	75%	13%	87%
Belimumab (healthy)	77%	10%	90%
Belimumab (patient)	68%	9%	91%
Canakinumab	68%	19%	81%
Daclizumab	57%	19%	81%
Golimumab	84%	9%	91%
Guselkumab	35%	24%	76%
Trastuzumab	65%	17%	83%

that among these ten mAbs, the calibrated value of k_{12} exhibits a weak negative correlation with k_{23} while demonstrating a moderate positive correlation with k_{42}. Additionally, k_{34} displays a strong positive correlation with k_{23}. Therefore, if a mAb, referred to as "A", has a higher value of k_{23} than another mAb, referred to as "B", then mAb "A" is likely to have a greater value of k_{34} than mAb "B".

It should be noted that the reliability of predicted mean and standard deviations obtained from Bayesian inference depends on the quality and quantity of available data, the model assumptions, and the prior distribution, which encodes information about calibrated parameters before calibration. In Bayesian inference, the prior distribution is combined with the likelihood function to form the posterior distribution (Gelman, 2002). When data is scarce, the prior distribution can significantly influence the posterior estimates, potentially leading to biased results if the prior information is not properly chosen. However, as the dataset grows larger, the impact of the choice of prior distributions on posterior inferences decreases, and consequently, the posterior distribution becomes more tightly constrained around the true parameter values.

5. Results

Given variabilities in the parameters k_{12}, k_{23}, k_{34}, k_{42}, k_a, k_r, k_d, k_{diss}, and m_{F0} for subcutaneously administered therapeutic mAbs, we aim to explore the distribution of the pharmacokinetic parameters and investigate their correlations. Additionally, we intend to identify the factors that significantly affects these pharmacokinetic parameters.

5.1. Distribution of pharmacokinetic parameters of mAbs

We are interested in studying the time (T_{max}) taken to reach the peak concentration of mAbs in the plasma, the half-life ($T_{1/2}$) of mAbs in the plasma, and the bioavailability. The variability of rate constants in our compartment model depends on factors such as injection parameters, tissue properties, and mAb characteristics. Unfortunately, this information is currently unavailable. To facilitate our analysis, we assume that the parameters k_{12}, k_{23}, k_{34}, k_{42}, k_a, k_r, k_d, k_{diss}, and m_{F0} follow a uniform distribution within the ranges specified in Table 5. The predicted mean of these parameters is the same as that for administering 200 mg of belimumab to healthy volunteers, as listed in Table 2, but these parameters have 20% variability around their expected values.

Using Latin hypercube design (McKay et al., 1979), we randomly sample 50,000 points in the parameter space. Subsequently, we substitute the values of k_{12}, k_{23}, k_{34}, k_{42}, k_a, k_r, k_d, k_{diss}, and m_{F0} from these 50,000 sampling points into our compartment model and obtain the pharmacokinetic parameters T_{max}, $T_{1/2}$, and the bioavailability. The distribution of the pharmacokinetic parameters is shown in Fig. 6. The skewnesses, which measures the asymmetry of a distribution, for T_{max} and $T_{1/2}$ are 0.22 and 0.45, respectively, indicating that their mean values are greater than their medians. Conversely, the skewness of bioavailability is −0.13, indicating that the mean value for bioavailability is less than its median. Additionally, the skewness for T_{max}, $T_{1/2}$, and bioavailability is between −0.5 and 0.5, illustrating that their distributions are approximately symmetric. Therefore, reporting standard statistical inferences such as a confidence interval (e.g., 95%) with a mean for these pharmacokinetic parameters is reasonable.

5.2. Correlation analysis

For the previously considered 50,000 data points, Fig. 7 shows the correlations among the pharmacokinetic parameters of mAbs and the rate constants k_{12}, k_{23}, and k_a. The variables k_{12}, k_{23}, and k_a are assumed to be independent when we sample the 50,000 data points; hence, their correlations are close to zero.

The rate constant k_{12} exhibits a weak positive correlation with bioavailability and a moderate negative correlation with T_{max}. The rate constant k_{23} shows a moderate positive correlation with bioavailability and a weak negative correlation with T_{max}. Additionally, k_{23} displays a moderate positive correlation with $T_{1/2}$, while k_{12} shows only a negligible correlation with $T_{1/2}$. Furthermore, the rate of cellular uptake (k_a) is negatively correlated with T_{max}, $T_{1/2}$, and bioavailability. We observe a strong positive correlation between $T_{1/2}$ and bioavailability, along with a weak positive correlation between $T_{1/2}$ and T_{max}. However, T_{max} demonstrates a negligible correlation with bioavailability.

5.3. Parametric studies

Fig. 8(a) illustrates a reduction in T_{max} as k_{12}/k_a increases. Additionally, the impact of k_{12}/k_a on T_{max} is more pronounced than that of k_{23}/k_a unless k_{23}/k_a is significantly smaller than k_{12}/k_a. In contrast, Fig. 8(b) reveals that k_{23}/k_a has a greater effect on $T_{1/2}$ compared to k_{12}/k_a; $T_{1/2}$ increases with the growth of k_{23}/k_a. The bioavailability increases as the values of k_{12}/k_a and k_{23}/k_a rise, as shown in Fig. 8(c). Nevertheless, as the values of k_{12}/k_a and k_{23}/k_a increase, their impact on bioavailability decreases since the contour lines become looser. Therefore, if k_{12}/k_a is considerably smaller than k_{23}/k_a, adjusting k_{12}/k_a is expected to be more effective in enhancing bioavailability, and vice versa.

Table 5
The ranges considered for the parameters k_{12}, k_{23}, k_{34}, k_{42}, k_a, k_r, k_d, k_{diss}, and m_{F0}.

Parameter	k_{12}	k_{23}	k_{34}	k_{42}	k_a	k_r	k_d	k_{diss}	m_{F0}
Range	(1.16, 1.75)	(0.71, 1.06)	(0.36, 0.54)	(3.83, 5.75)	(0.13, 0.19)	(0.66, 1)	(0.82, 1.24)	(0.14, 0.21)	(0.05, 0.07)

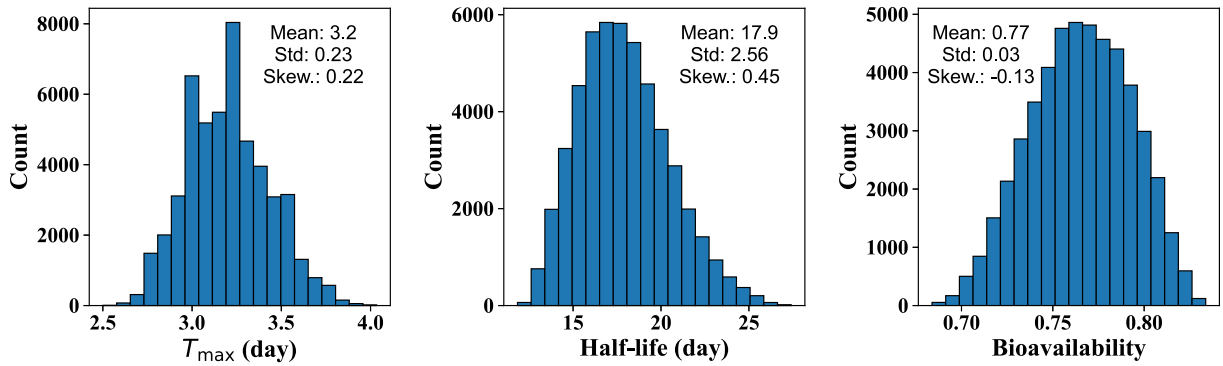

Fig. 6. Distribution of the T_{max}, $T_{1/2}$, and bioavailability for the SC injection of belimumab into healthy volunteers.

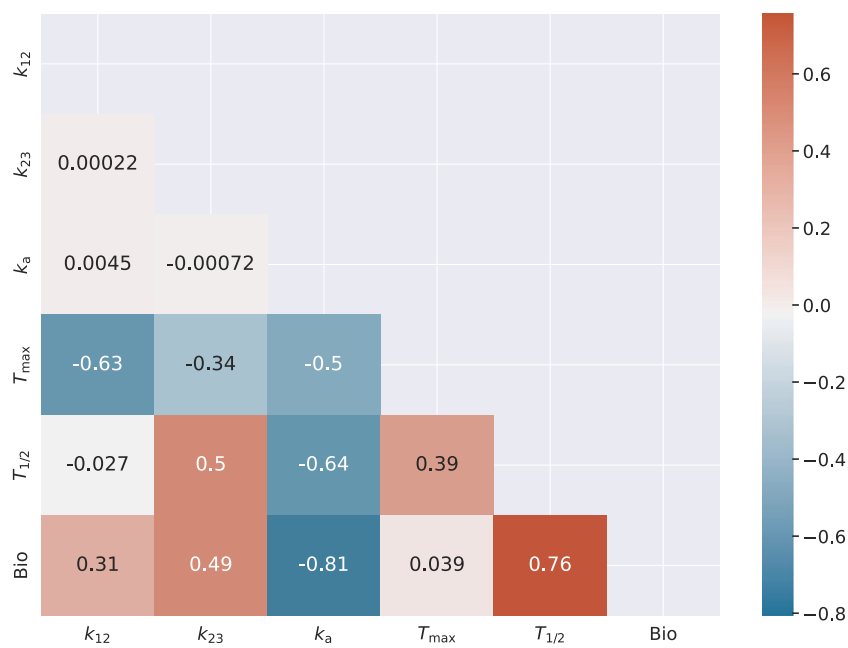

Fig. 7. The correlation among T_{max}, $T_{1/2}$, bioavailability, and the rate constants k_{12}, k_{23}, and k_a.

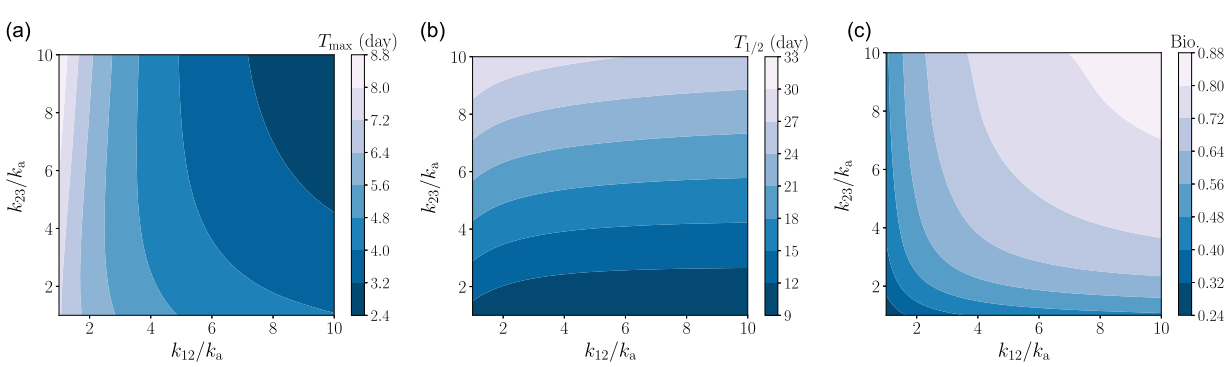

Fig. 8. The phase plot quantifying the impact of k_{12}/k_a and k_{23}/k_a on T_{max}, $T_{1/2}$, and bioavailability.

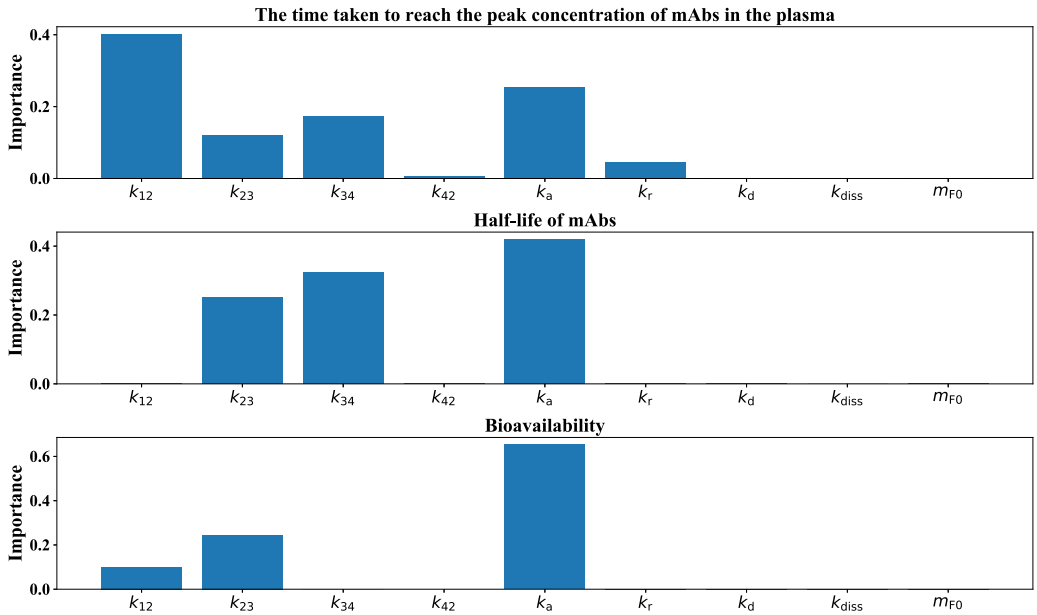

Fig. 9. Sobol Sensitivity analysis for the pharmacokinetic parameters of subcutaneously administered belimumab into healthy volunteers.

5.4. Sensitivity analysis

The Sobol sensitivity analysis is a powerful tool for identifying factors that significantly influence the pharmacokinetics of mAbs following SC injection (Zhang et al., 2015). It not only provides guidance for future studies but also enables a better understanding of the impact of uncertainties in the rate of transport of mAbs between adjacent compartments on their pharmacokinetic parameters. The basic idea of the Sobol sensitivity analysis is to decompose the output variance into components that represent the variances of the input parameters at progressively higher levels of dimensionality (Saltelli et al., 2004). Specifically, if our input parameters are denoted as $\Theta = \{\theta_1, \theta_2, \dots, \theta_s\}$, and each input parameter is considered to be uniformly distributed over some finite interval. Our first step is to rescale the range to [0,1] and use Sobol sequence to generate uniform sampling in the $s-$dimensional unit hypercube. Next, we decompose the variance of the model output (D) as:

$$D = \sum_{i=1}^{s} D_i + \sum_{i<j} D_{ij} + \sum_{i<j<l} D_{ijl} + \cdots + D_{1,2,3,\dots,s} \tag{14}$$

where $D_{p_1,\dots,p_k}$ $(1 \le k \le s)$ is the partial variance corresponding to the subset of parameters $\{p_1, \dots, p_k\}$. The detailed calculation for $D_{p_1,\dots,p_k}$ can be found in the literature (Zhang et al., 2015). After the decomposition, we can compute the total-order sensitivity index (S_{Ti}), which quantifies the overall effects of the parameter θ_i on the output variance:

$$S_{Ti} = S_i + \sum_{j \neq i} S_{ij} + \cdots + S_{1,2,\dots,i,\dots,s} \tag{15}$$

where $S_i = D_i/D$ is the first-order contribution from the input parameter θ_i to the output variance, and $S_{ij} = D_{ij}/D$ is the second-order contribution from interaction between θ_i and θ_j, and so on.

We use the open-source library, SALib (Herman and Usher, 2017), to perform the Sobol sensitivity analysis, and we report the total-order sensitivity index (S_{Ti}) in Fig. 9. Similarly, for the other nine mAbs included in the human study, we assume that they share the same predicted means as those listed in Tables 1–3, with a 20% variability around their predicted means. The results of the sensitivity analysis for these nine mAbs are included in the supplementary material. Our sensitivity analyses for these ten mAbs reveal that (1) k_{12}, k_{34}, and k_a significantly affect T_{max}, (2) k_{23}, k_{34}, and k_a significantly influence $T_{1/2}$, and (3) k_{12}, k_{23}, and k_a significantly impact bioavailability.

We also perform a sensitivity analysis on rat data, where rituximab is administered via a SC route with a mass of 0.38 mg. The corresponding results are shown in Figure S11 within the supplementary material. Similarly to the trends observed in human data, the effect of k_{12} on T_{max} remains more pronounced than that of k_{23}. Additionally, k_{23} exhibits a more significant effect than k_{12} on $T_{1/2}$. Moreover, our calibrated value of 1 mg for the mass of FcRn in the fifth and sixth compartments is comparable to the mass of the administered rituximab, the activities within these compartments gain significance and impact T_{max}, $T_{1/2}$, and bioavailability.

6. Discussion

Our sensitivity analysis indicates that bioavailability increases with the rate constants k_{12} and k_{23}, potentially enhancing our understanding of certain clinical observations and enabling predictive insights. For instance, experimental studies have demonstrated that therapeutic proteins with a higher charge and stronger hydrophobic interactions with the ECM tend to exhibit lower bioavailability (Datta-Mannan et al., 2020). This decrease in bioavailability is likely attributed to a reduction in k_{12}. Additionally, experimental studies have revealed that an increase in body weight corresponds to a decrease in the pumping frequency of collecting lymphatics and the initial lymphatic density (Nitti et al., 2016). This reduction in initial lymphatic density and pumping frequency of collecting lymphatics results in decreased lymphatic uptake rate and lymph flow rate (Weitman et al., 2013), respectively, contributing to the decrease of bioavailability, consistent with the literature finding (Hanzel et al., 2021) that the heavy patients had lower cumulative drug exposure. However, the body weight also affects other parameters. For instance, as the body weight increases, the thickness of the SC layer and the clearance of mAbs increase (Gill et al., 2016b; Thomas and Balthasar, 2019; Bensalem and Ternant, 2020). Thus, additional experimental and clinical studies are needed to confirm the above predictions.

Despite accounting for numerous factors that influence the pharmacokinetics of therapeutic mAbs, our compartment model has some limitations. First, we do not consider pharmacodynamics and immunogenicity, which are important properties of therapeutic mAbs that may affect their efficacy and safety (Crommelin and Sindelar, 2008; Wang et al., 2017). In future studies, we can incorporate pharmacodynamics models, such as the sigmoid E_{max} model (Salahudeen and Nishtala,

2017), cell lifespan models (Woo and Jusko, 2007), and indirect pharmacodynamic response models (Pan et al., 2019), to better understand the interplay between the concentration of therapeutic mAbs and the desired therapeutic effect.

Second, our model does not account for the flux of mAbs from the interstitial fluid to the blood vessel due to the tight junctions between vascular endothelial cells. However, this applies to targets under normal conditions, and inflammation may alter the intercellular gap junctions between vascular endothelial cells, increasing vascular permeability (Triacca et al., 2017).

Third, a clear relationship between the rate constants and physiological parameters is currently missing. Addressing this issue in future studies can guide the design of injection devices (Zhong et al., 2021, 2022) and the selection of appropriate injection sites, depths, and rates. To address this issue, we need to integrate our compartment model with high-fidelity numerical simulations to explore the effects of injection parameters, mAb properties, and tissue characteristics on the pharmacokinetic parameters of therapeutic mAbs (Li et al., 2023; Rahimi et al., 2022). This work could potentially address concerns related to species differences, provided that the transport of mAbs described by our compartment model holds true for both rodents and humans, despite differences in their rate constants and FcRn quantities.

Fourth, our compartment model is based on multiple assumptions and simplifications, and their validity needs justification. For instance, we assume that the amount of FcRn in the vascular space and periphery tissues is sufficient compared to the administered mAbs, while the amount of FcRn in the local SC tissue and lymphatic system is limited and equal. As our understanding of the physiological parameters and the mAb transport process deepens, the new knowledge should be incorporated into the compartment model to improve its predicted capabilities.

Fifth, the accuracy of Bayesian calibration results depends on the quality and quantity of experimental data. As the available rat and human data are limited, especially the human data considered in this manuscript exhibits linearity to doses, our calibration results may have biases and uncertainties. Additionally, different combinations of calibrated values may yield similar predicted results, which can limit the interpretability of the calibrated values. If more experimental data is available in the future, our calibration results will become more reliable. In that case, we could use our calibrated results to gain insights into inter-individual variability and account for the species differences. Furthermore, if we have a large dataset or have performed measurements to reduce the number of calibrated parameters in the future, simultaneous calibration can be considered to distinguish between parameters specific to the mAbs and those specific to the system.

7. Conclusion

In this paper, we have developed a compartmental model that distinguishes between FcRn-expressing cells at different sites. This model has been validated against the experiments of human and rat studies. Our correlation analysis reveals that T_{max} has a weak positive correlation with the half-life of mAbs and a negligible correlation with bioavailability. In contrast, the half-life of mAbs exhibits a strong positive correlation with bioavailability.

Additionally, our sensitivity analysis indicates that the rate of mAb transport within the local subcutaneous tissue, denoted by k_{12}, significantly influences T_{max}, whereas the rate of mAb transport within the lymphatic system, represented by k_{23}, significantly affects the half-life of mAbs. Moreover, bioavailability is impacted by the transport of mAbs within both the interstitial space and the lymphatic system. Consequently, the impact of injection parameters on the rate of lymphatic uptake may or may not translate into changes in bioavailability, depending on the relative importance of k_{12} and k_{23}.

This study has quantified the significance of the rate constants of therapeutic mAbs on their pharmacokinetic parameters. Additionally, integrating available simulations for mAb transport within the interstitial space with this compartmental model could further investigate the impact of injection parameters on the pharmacokinetic parameters of therapeutic mAbs. Industries can effectively employ this compartment model to develop strategies aimed at enhancing the bioavailability of therapeutic proteins.

CRediT authorship contribution statement

Xiaoxu Zhong: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **Yikai Liu:** Data curation, Validation. **Arezoo M. Ardekani:** Conceptualization, Formal analysis, Investigation, Methodology, Funding acquisition, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

XZ and YL are research assistants, AMA is a professor in Mechanical Engineering at Purdue University.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijpharm.2023.123687.

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